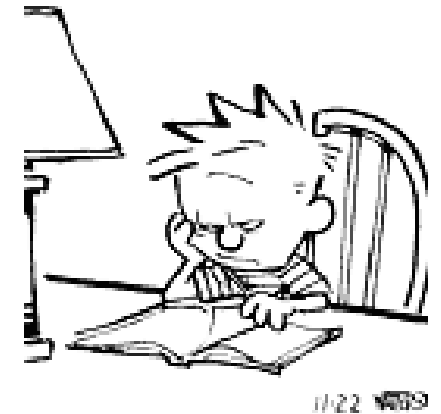


Project guidelines

Rémi Veltin – 2026-01

Guidelines

- **Project Objective:** The aim is to **present a paper** to the class that involves **supervised, unsupervised or deep learning**. Participants should be able to **replicate the study's findings on a smaller scale**. This exercise is designed to introduce new concepts, methods, and models to the class, as well as to enhance Python programming skills.
- **Data Selection:** Data associated with the paper will often be your primary source. If the data is not publicly available, you may need to find alternative sources or generate the data yourself. It is advisable to verify the availability of data before proceeding. Please submit the paper to the teacher for approval and further discussion.
- **Task Type:** Supervised, Unsupervised ML, DL or a combination.
- **Groups:** This project is to be completed in groups.
- **Evaluation:** The project will be evaluated through **a presentation (.pptx like)** and **a Streamlit demo**, lasting approximately 10-20 minutes depending on the number of groups, including questions.
- **Deliverables:**
 - > 1 presentation = explain the content of the paper and how you reproduce it (~20 slides)
 - > 1 python package .zip = back-end (data pipeline, model,...) and front-end (Streamlit)
- **Tips:** Ensure you clearly explain your project and the data flow. This includes:
 1. Collection and acquisition of data
 2. Data processing and preparation
 3. Model selection and optimization
 4. Performance evaluation and the choice of scoring metrics



Follow the best practices for data projects to ensure success !

ML stages → key steps

1. Look at the big picture.
2. Get the data.
3. Discover and visualize the data.
4. Prepare the data for Machine Learning algorithms.
5. Select a model and train it.
6. Fine-tune your model(s)
7. Present your solution.
8. Launch, monitor, and maintain your system.



ML Stages DALL-E generated

Let's go back to ML stages → ML flow

